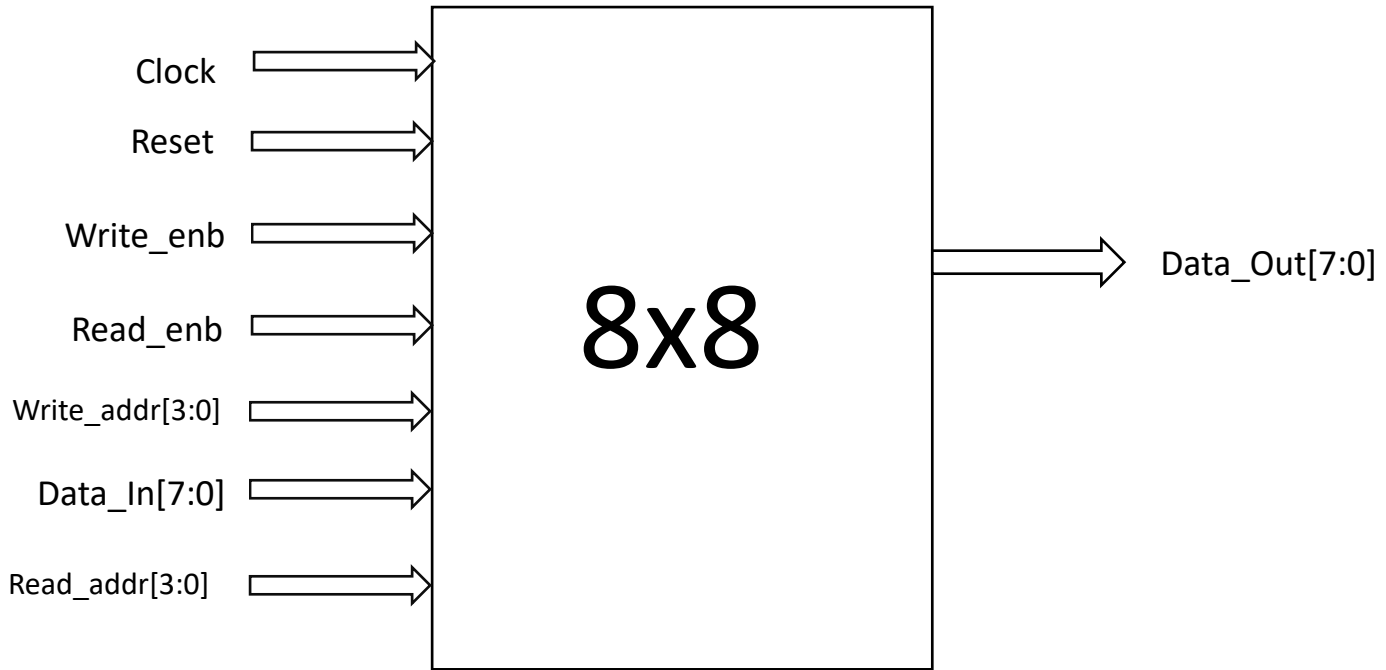


BLOCK DIAGRAM AND FUNCTIONING OF 8x8 RAM



- When Write_enb = 1; → We can write address location by Write_addr.
- Whereas Write_enb = 0; → write_addr are not passed.
- Similarly in Read_enb = 1; → We can read our data by registered address sent from Write_addr.
- In Read_enb = 0; → Write_addr was working and passing datas.
- Data_in[7:0] is sent in → Write_enb = 1; Write_addr → written address.
- Clock → Whole circuit synchronous with posedge (except reset).
- Reset → It is Asynchronous (not followed by clock).
- Data_out[7:0] is received when → Read_en = 1; Read_addr → address allocated by write_addr.

Example

// In input,

write_enb = 1 → write_addr = 3'd5; → Data_in = 8'b 00110001 (data sent in specific address).

After some time, in positive edge of clock,

Read_enb = 1 → Read_addr = 3'd5; → Data_out = 8'b 00110001 (data received as mentioned).